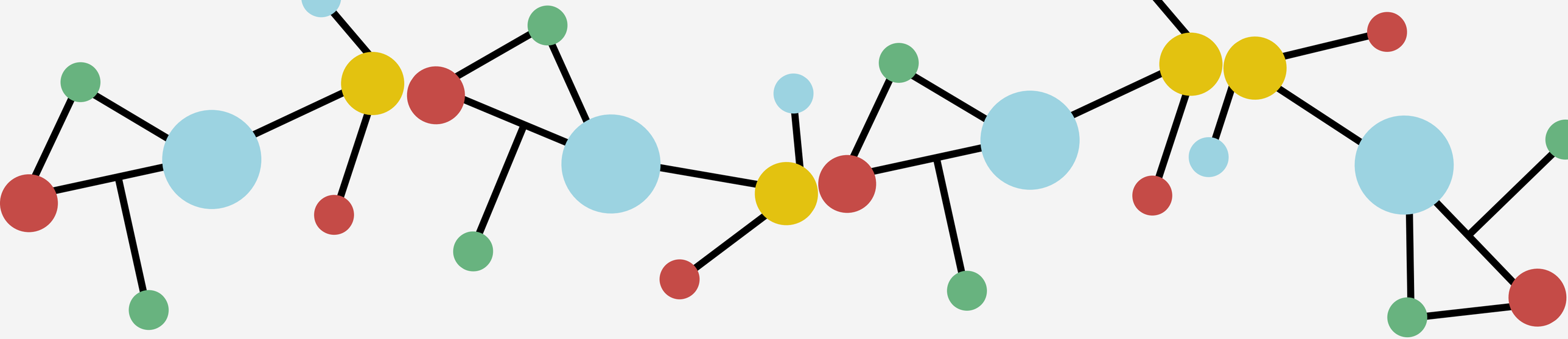


DS-UNIMIB Knowledge Graph

Sara Borello

Keita Jacopo Viganò

The Idea

This project focuses on constructing a graph database to map the professors of the course of Data Science of University of Milan Bicocca



Trend Analysis

Identify the specific research areas and topics explored by professors, supporting students in selecting thesis supervisors aligned with their interests.



Collaboration Networks

Mapping connections between researchers and their co-authors to reveal patterns of cooperation.



Affiliation Insights

Highlighting partnerships and affiliations with external institutions to understand the department's global impact.

Data and Sources

DATA

AUTHORS:

- https://info.orcid.org/what-is-orcid/services/public-api/
- https://docs.openalex.org/how-to-use-the-api/api-overview

PAPERS:

- https://www.crossref.org/documentation/retrieve-metadata/rest-api/
- https://docs.openalex.org/how-to-use-the-api/api-overview

COURSES:

- Data science Regulation

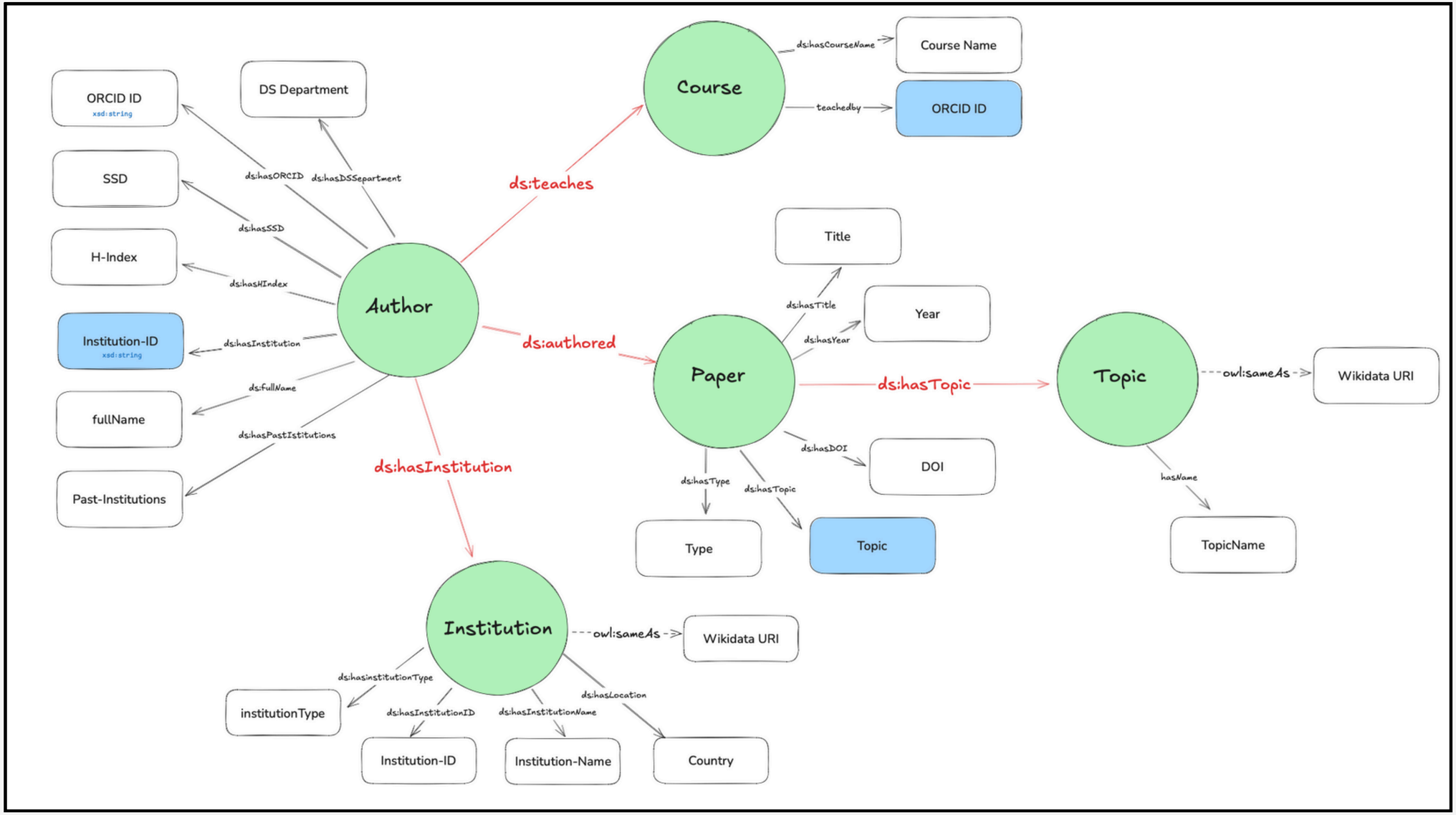
TOPICS & INSTITUTIONS:

- https://docs.openalex.org/how-to-use-the-api/api-overview

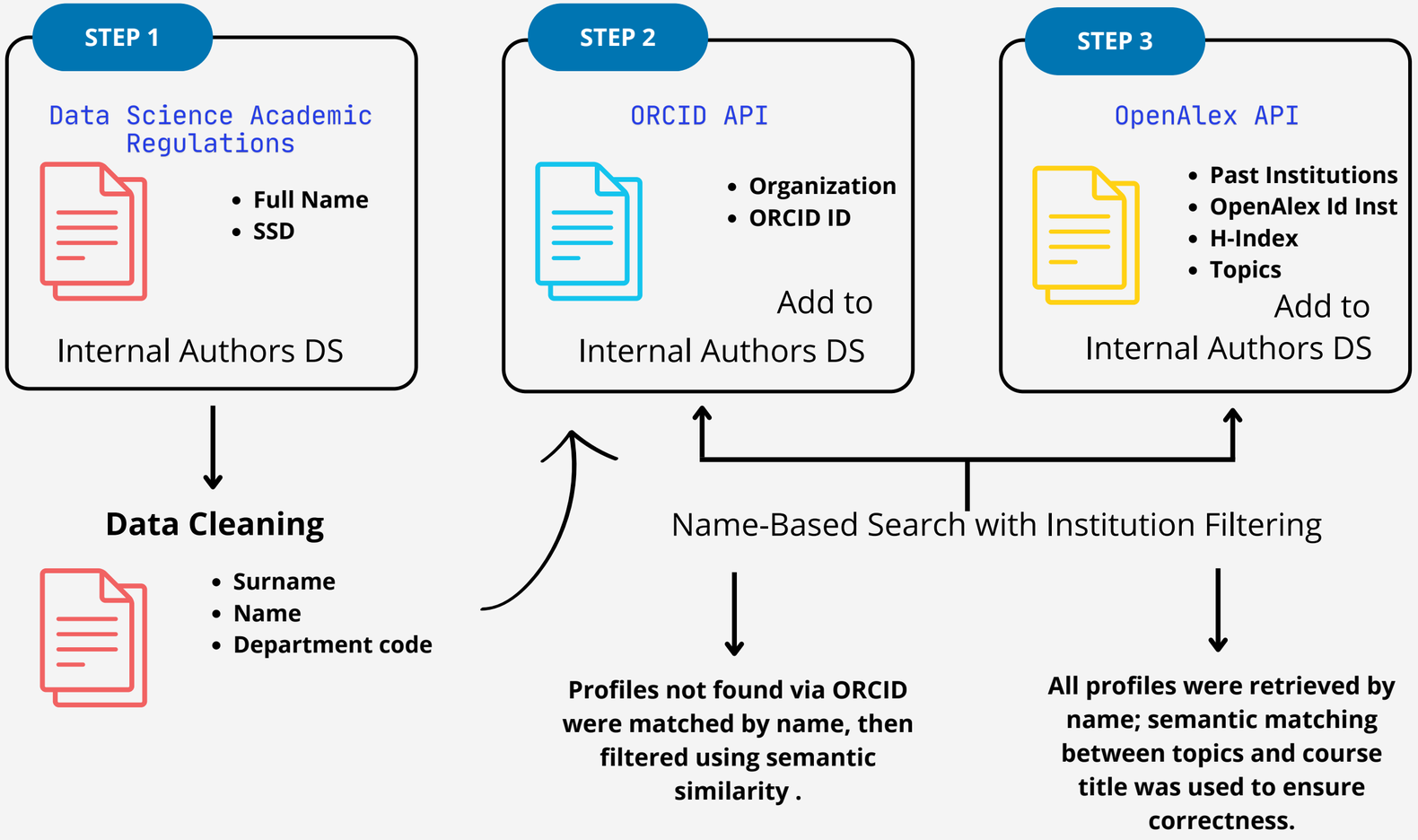
SOURCES

- Angioni, S., Salatino, A. A., Osborne, F., R- eforgiato Recupero, D., & Motta, E. (2022). AIDA: A knowledge graph about research dynamics in academia and industry. Quantitative Science Studies, 2(4), 1356–1398. https://doi.org/10.1162/qss_a_00162

Conceptual RDF Schema of the Knowledge Graph



Data Acquisition: Authors Internal



Semantic Matching: ORCID Profiles

Candidate text string

'DAVIDE PAOLO BERNASCONI How Are Diagnosis-Related Groups and Staffing Allocation Systems Associated with the Complexity of Nursing Care? An Observational Study; Ultrasound-Assisted, Catheter-Directed Thrombolysis for Acute Intermediate/High-Risk Pulmonary Embolism: Design of the Multicenter USAT IH-PE Registry and Preliminary Results; The Role of Non-Peripancreatic Lymph Nodes in the Survival of Patients Suffering from Pancreatic....

"Davide Paolo Bernasconi Capsoni, N., Azin, G., Scarnera, M., Bettina, M., Breviario, R., Ferrari, L., et al. (2025). Bloodstream infections due to multi-drug resistant bacteria in the emergency department: prevalence, risk factors and outcomes—a retrospective observational study. INTERNAL AND EMERGENCY MEDICINE, 20(2), 573-583 [10.1007/s11739-024-03692-7]. Dettaglio; Famularo, S., Milana, F., Ardito, F., Cipriani, F., Vitale, A., Lauterio, A., et al. (2025).....

Professor text string

- Each professor is represented through a text string combining name, surname, and publication titles.
- Candidate ORCID profiles with the same name/surname are processed similarly.

Candidate embeddings

tensor([7.5331e-02, -8.0090e-02, -6.6386e-03, -4.0188e-02, -1.1406e-02, -2.3928e-02, -6.7087e-02, 9.5181e-02, 1.6072e-02, -2.8959e-02, ..

tensor([1.9537e-02, -1.5883e-02, -9.9302e-02, -6.4353e-02, -1.5682e-02, -3.4050e-02, -1.2380e-01, 1.2724e-01, 6.9647e-03, 2.4253e-03,...

Professor embeddings

- Text strings are encoded using the SentenceTransformer model "**all-MiniLM-L6-v2**" to obtain semantic embeddings.

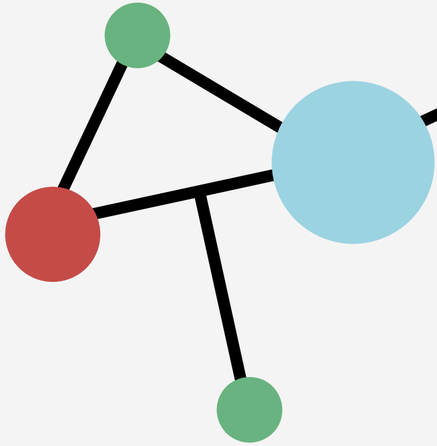
$$\text{cosine_similarity}(\mathbf{A}, \mathbf{B}) = \frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\| \cdot \|\mathbf{B}\|}$$

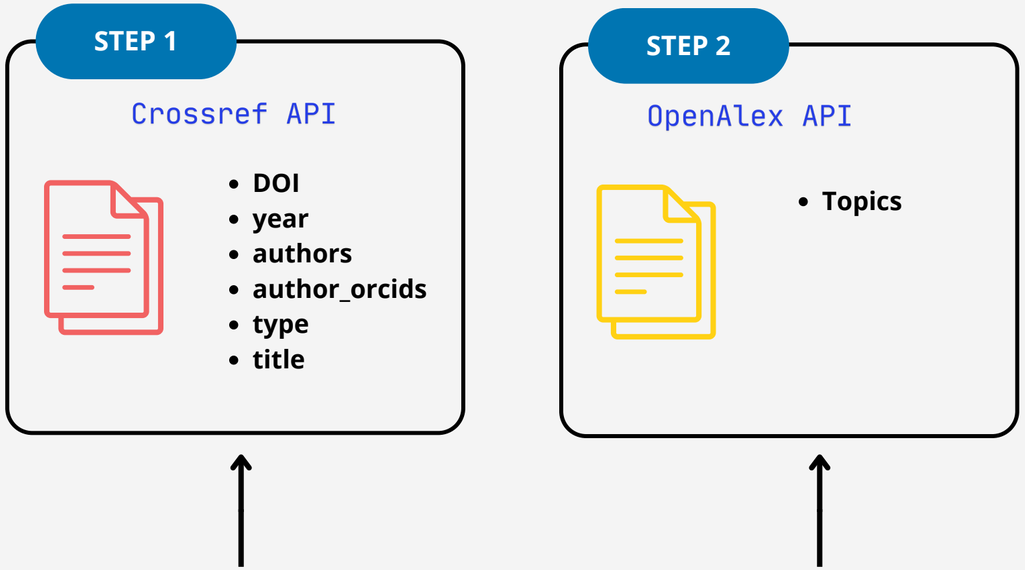
Similarity Score: 0.321503

- The cosine similarity between the professor's vector and each ORCID candidate's vector is computed.
- The ORCID profile with the highest similarity score is selected as the best match.
- Final output links each professor to their most likely ORCID ID, supported by a quantitative similarity score.

Same approach used between topics and course title to ensure correctness.

Data Acquisition: Papers



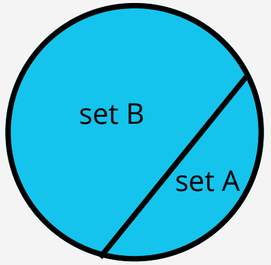




Data cleaning: Remove duplicate and inconsistent papers > 10 Authors and < 2018 of publishing's year

Data Acquisition: Authors External

set B: Unique ORCIDs of external authors



set A: Unique ORCIDs of internal authors derived from **Internal Authors Dataset**

Only **set B ORCIDs**

From Papers

Authors Unique Column composed by ORCIDs

STEP 1

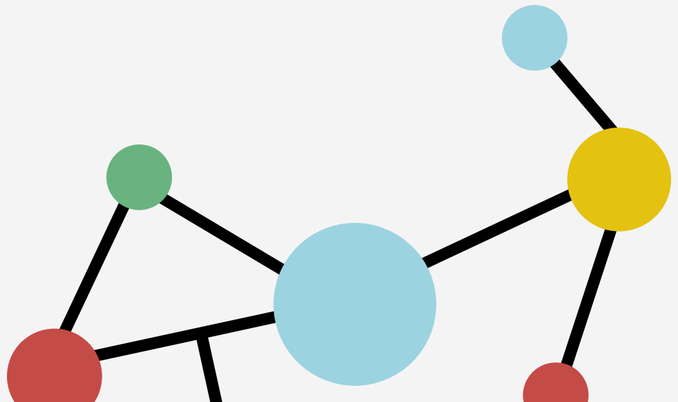
`OpenAlex API`

Search by ORCID

- **Name**
- **Openalex Id**
- **Institution ID**
- **Institution Name**
- **Institution type**
- **Institution Country**
- **H index**

Add to External Authors DS

Dropped in Institution



Data Acquisition: Institution

STEP 1

From Authors External

Unique

- **Institution ID**
- **Institution Name**
- **Institution type**
- **Institution Country**

Add to
Institution

STEP 2

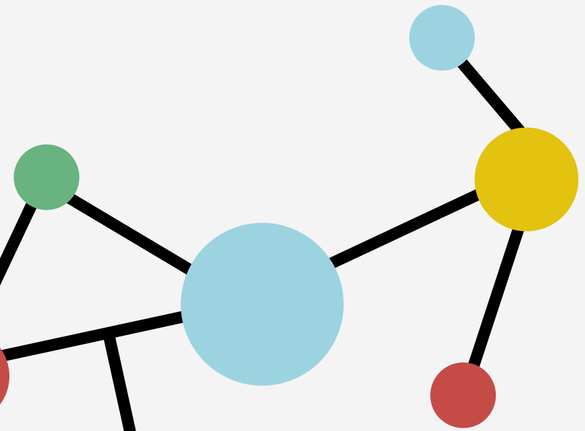
Openalex

- **Wikidata URI**

Add to
Institution

Wikidata URI to enrich each Institution node with external knowledge:

- Enables semantic expansion via federated SPARQL queries
- Connects to curated data (type, country, network, etc.)
- Improves interoperability across datasets
- Supports reasoning and classification tasks



Data Acquisition: Topic

STEP 1

From Paper

Unique

- **Topic**

Add to Topic

STEP 2

Openalex

- **Wikidata URI**

Add to Topic

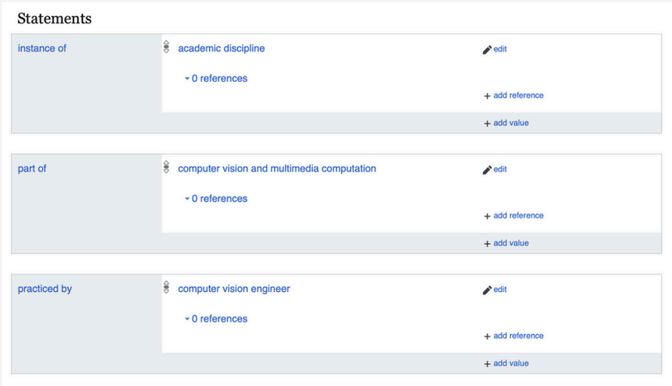
Wikidata URI to enrich each Institution node with external knowledge:

- Enables semantic expansion via federated SPARQL queries
- Connects to curated data (type, country, network, etc.)
- Improves interoperability across datasets
- Supports reasoning and classification tasks

EXAMPLE

Computer Vision and Pattern Recognition

https://www.wikidata.org/wiki/Q844240



Ontology Design and RDF Modeling

Technology stack

RDF

Structuring data as subject–predicate–object triples

RDFS

Defining schema-level semantics: class hierarchies (rdfs:subClassOf), property domains and ranges

OWL

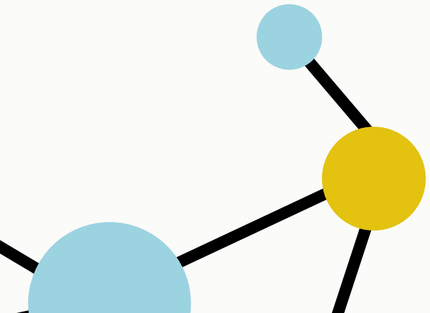
Adding expressive logic: inverse properties (owl:inverseOf), external linking (owl:sameAs), and property chains (owl:propertyChainAxiom)

RDF4J

Although the ontology was created programmatically using rdflib in Python, it was exported as a Turtle (.ttl) file and subsequently imported into RDF4J for storage, querying (via SPARQL), and reasoning.

OWL RL reasoner

The OWL RL reasoner in Python (owlrl) infers implicit facts in an RDF graph using rule-based OWL RL semantics.



Ontology Design and RDF Modeling

TBox Formalization: OWL Classes and Logical Axioms

OWL Classes

Class	Description
`ds:Author`	A researcher affiliated with the department
`ds:Course`	A course taught in the department
`ds:Paper`	A scientific publication authored by researchers
`ds:Topic`	Research topic, aligned with Wikidata concepts
`ds:Institution`	Academic or research institution
`ds:Person`	Superclass of `ds:Author`

TBox contains schematic knowledge: OWL classes, object and data properties, and relationships between concepts expressed through logical axioms.

TBox Axioms in Description Logic

$\text{Author} \sqsubseteq \text{Person}$ — **RDFS.subClassOf**

$\text{Author} \sqcap \text{Paper} \sqsubseteq \bot$
$\text{Author} \sqcap \text{Course} \sqsubseteq \bot$
$\text{Paper} \sqcap \text{Course} \sqsubseteq \bot$ — **OWL.disjointWith**

$\exists \text{authored}^{-} \sqsubseteq \text{Paper}$ — **RDFS.range**
$\exists \text{authored} \sqsubseteq \text{Author}$ — **RDFS.domain**
$\text{authored} \sqsubseteq \top \quad (\text{dom} = \text{Author}, \text{range} = \text{Paper})$
OWL.ObjectProperty

$\exists \text{hasAuthor}^{-} \sqsubseteq \text{Author}$ — **RDFS.range**
$\exists \text{hasAuthor} \sqsubseteq \text{Paper}$ — **RDFS.domain**
$\text{hasAuthor} \sqsubseteq \top \quad (\text{dom} = \text{Paper}, \text{range} = \text{Author})$
OWL.ObjectProperty

OWL.ObjectProperty
$\text{hasTopic} \sqsubseteq \top \quad (\text{dom} = \text{Paper}, \text{range} = \text{Topic})$
$\text{teaches} \sqsubseteq \top \quad (\text{dom} = \text{Author}, \text{range} = \text{Course})$
$\text{hasInstitution} \sqsubseteq \top \quad (\text{dom} = \text{Author}, \text{range} = \text{Institution})$

$\text{Functional}(\text{hasORCID}) \quad \text{hasORCID}: \text{Author} \to \text{xsd:string}$
$\text{Functional}(\text{hasDOI}) \quad \text{hasDOI}: \text{Paper} \to \text{xsd:string}$
OWL.FunctionalProperty

Ontology Design and RDF Modeling

ABox Layer in the Ontology: Populating Knowledge with Real-World Assertions

Overview of ABox Instances and Associated Object/Data Properties

Entity	Example Individual	All Properties (Object + Data)
Author	a_1	hasORCID, fullName, hasSSD, hasHIndex, hasOpenAlexID, pastInstitutionsID, hasDSDepartment, hasInstitution, authored, teaches, rdfs:label
Institution	i_1	institutionName, institutionType, institutionCountry, owl:sameAs, rdfs:label
Course	c_1	courseName, rdfs:label
Paper	p_1	hasDOI, hasTitle, hasYear, hasType, hasAuthor, hasTopic, rdfs:label
Topic	t_1	rdfs:label, owl:sameAs

The ABox contains factual statements about individual entities (authors, papers, courses, topics, and institutions) instantiated from the TBox vocabulary. It reflects real-world data from CSV sources, encoded as RDF triples.

ABox Instantiations in OWL Manchester Syntax

```
Individual: a1
    Types: ds:Author
    Facts:
        ds:hasORCID "0000-0001-2345-6789",
        ds:fullName "Alice Rossi",
        ds:hasHIndex 12,
        ds:hasSSD "INF/01",
        ds:hasInstitution i1,
        ds:teaches c1 .....

Individual: i1
    Types: ds:Institution
    Facts:
        ds:institutionName "University of Milano-Bicocca",
        ds:institutionType "University",
        ds:institutionCountry "Italy",
        owl:sameAs <http://www.wikidata.org/entity/Q133813>

Individual: c1
    Types: ds:Course
    Facts:
        ds:courseName "Data Science and Society",
        rdfs:label "Data Science and Society"

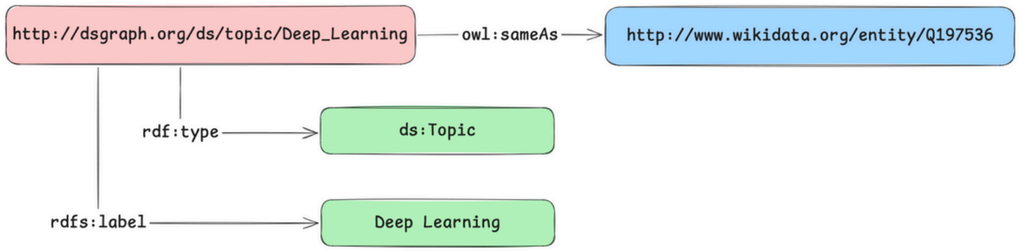
Individual: p1
    Types: ds:Paper
    Facts:
        ds:hasDOI "10.1234/abc123",
        ds:hasTitle "Knowledge Graph Construction in Python",
        ds:hasYear 2024,
        ds:hasType "Conference Paper",
        ds:hasAuthor a1,
        ds:hasTopic t1

Individual: t1
    Types: ds:Topic
    Facts:
        rdfs:label "Semantic Web",
        owl:sameAs <http://www.wikidata.org/entity/Q54871>
```

Ontology Design and RDF Modeling

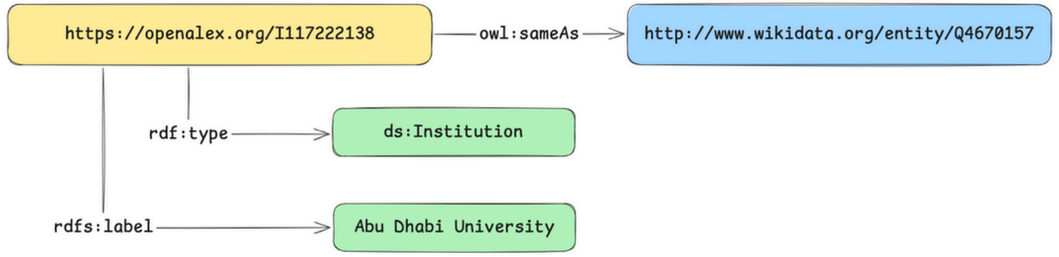
Semantic Linking: owl:sameAs is a property used to semantically align local entities with external identifiers from Wikidata

Topic Semantic Linking



The local concept ds:Deep_Learning is linked to its external equivalent in Wikidata through owl:sameAs, enabling semantic interoperability.

Institution Semantic Linking



The institution identified by its OpenAlex URI is typed as ds:Institution, labeled Abu Dhabi University, and linked to its external equivalent in Wikidata through owl:sameAs, enabling semantic interoperability.

owl:sameAs supports instance equivalence, allowing ontology alignment and knowledge fusion across heterogeneous data sources.

Ontology Design and RDF Modeling

The RBox defines the set of axioms concerning roles (properties) that represent relationships between individuals. It complements the TBox (which defines concepts/classes) and the ABox (which asserts facts about individuals).

The axiom **authored owl:inverseOf hasAuthor** establishes that the property authored is the inverse of hasAuthor, which in Description Logic (DL) is written as:

$$\texttt{authored} \equiv \texttt{hasAuthor}^{-}$$

This allows the ontology to infer that if an author is linked to a paper via hasAuthor, then the paper is also related to the author via authored.

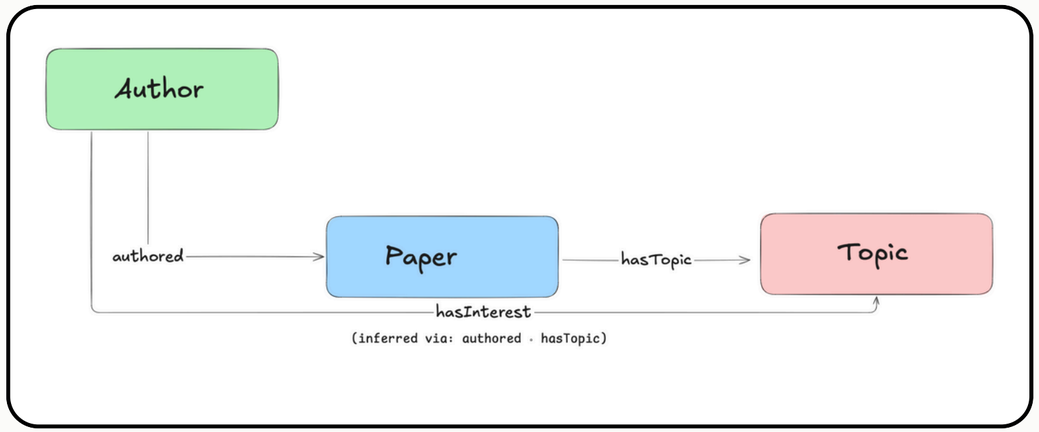
The constraint that every paper must have at least one author is modeled as a minimum cardinality restriction on the hasAuthor property with owl:Restriction and owl:minCardinality :

$$\texttt{Paper} \sqsubseteq \geq 1\ \texttt{hasAuthor}.\top$$

Ontology Design and RDF Modeling

The RBox defines the set of axioms concerning roles (properties) that represent relationships between individuals. It complements the TBox (which defines concepts/classes) and the ABox (which asserts facts about individuals).

Visual (OWL Property Chain)-hasInterest

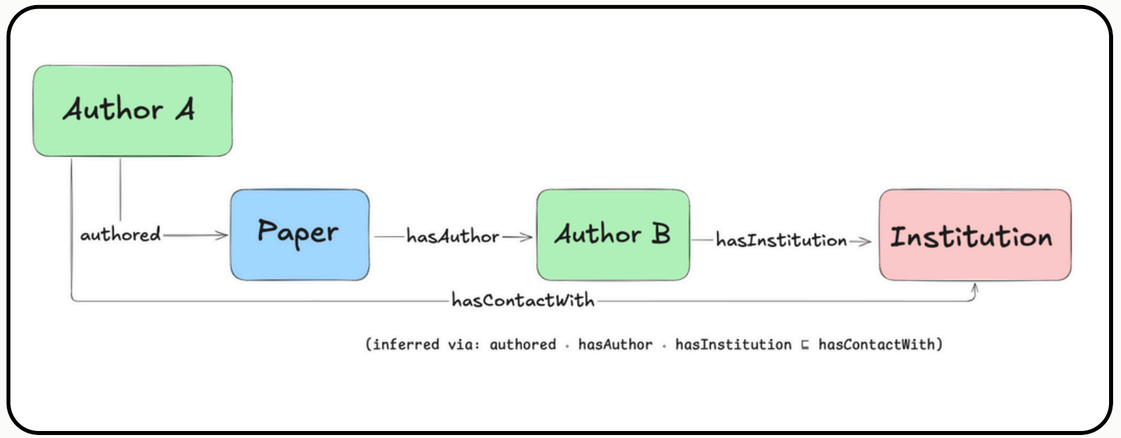


Semantics

If an Author **authored** a Paper and that Paper **hasTopic** a Topic, then the Author **hasInterest** in that Topic.

$$\text{authored} \circ \text{hasTopic} \sqsubseteq \text{hasInterest}$$

Visual (OWL Property Chain)-hasContact

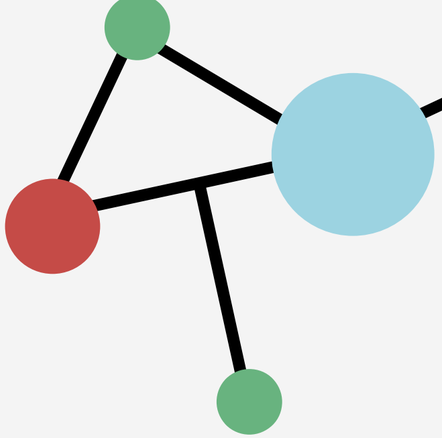


Semantics

If **Author A** has authored a **Paper**,
and that **Paper** has an **Author B**,
and **Author B** is affiliated with an **Institution**,
then **Author A** is inferred to have contact with that **Institution**.

$$\text{authored} \circ \text{hasAuthor} \circ \text{hasInstitution} \sqsubseteq \text{hasContactWith}$$

Data Querying



Topic Insights

Identifying the primary research areas and emerging topics within the professors of Data Science Course.

Research Question

Query

```
PREFIX ds: <http://dsgraph.org/ds#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>

SELECT ?topicLabel (COUNT(?paper) AS ?paperCount)
WHERE {
  ?author a ds:Author ;
          ds:authored ?paper .

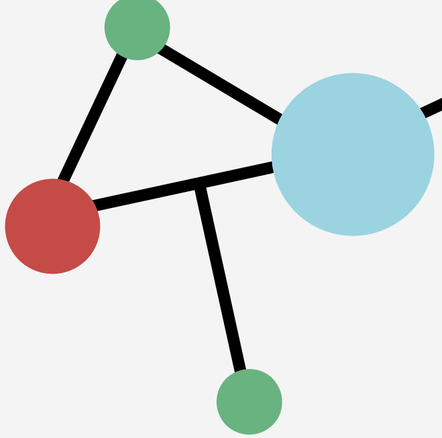
  ?paper ds:hasTopic ?topic .
  ?topic rdfs:label ?topicLabel .
}
GROUP BY ?topicLabel
ORDER BY DESC(?paperCount)

LIMIT 10
```

Result

```
Topic: Artificial Intelligence, Paper Count: 178
Topic: Molecular Biology, Paper Count: 78
Topic: Computer Vision and Pattern Recognition, Paper Count: 49
Topic: Information Systems, Paper Count: 43
Topic: Economics and Econometrics, Paper Count: 29
Topic: Sociology and Political Science, Paper Count: 28
Topic: Computer Networks and Communications, Paper Count: 24
Topic: Cardiology and Cardiovascular Medicine, Paper Count: 18
Topic: Endocrinology, Diabetes and Metabolism, Paper Count: 18
Topic: Management Science and Operations Research, Paper Count: 18
```

Data Querying



Affiliation Insights

Which institutions collaborate most frequently with the Professors of Data Science Science Course at the University of Milano-Bicocca, and what is the distribution of papers co-authored with these external institutions?

Research Question

```
PREFIX ds: <http://dsgraph.org/ds#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>

SELECT ?inst_name (COUNT(DISTINCT ?external_author) AS ?collab_count)
WHERE {
  # Author Internal (DS Department = 1)
  ?internal_author a ds:Author ;
                   ds:hasDSDepartment 1 ;
                   ds:authored ?paper .

  # External Authors of the same paper
  ?paper ds:hasAuthor ?external_author .
  FILTER (?external_author != ?internal_author)

  # Institution of external authors
  ?external_author ds:hasInstitution ?inst .
  ?inst ds:institutionName ?inst_name .
}
GROUP BY ?inst_name
ORDER BY DESC(?collab_count)
LIMIT 10
```

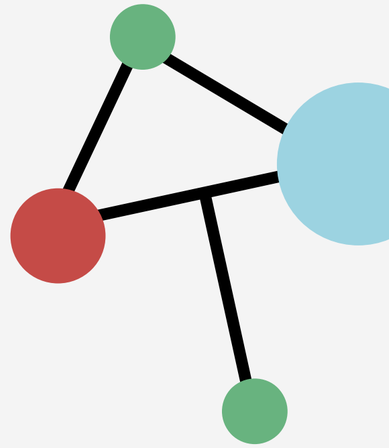
Query

```
University of Milano-Bicocca: 119
University of Milan: 13
Politecnico di Milano: 5
University of Padua: 5
Polytechnic University of Turin: 5
Fondazione Bruno Kessler: 5
Khalifa University of Science and Technology: 5
University of Rome Tor Vergata: 4
Istituti di Ricovero e Cura a Carattere Scientifico: 4
```

Result

Data Querying

Using Wikidata URI to expand Knowledge



Question

Retrieve the authors, titles, and topics of papers written by internal authors from the Data Science Course at the University of Milano-Bicocca, where the associated topics are linked to mathematical domain

Query

```
PREFIX ds: <http://dsgraph.org/ds#>
PREFIX owl: <http://www.w3.org/2002/07/owl#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX wdt: <http://www.wikidata.org/prop/direct/>
PREFIX wd: <http://www.wikidata.org/entity/>

SELECT DISTINCT ?author ?name ?paperTitle ?topicLabel
WHERE {
  # 1. Author inside Data Science Course
  ?author a ds:Author ;
          ds:fullName ?name ;
          ds:hasDSDepartment 1 ;
          ds:authored ?paper .

  # 2. Paper with topic
  ?paper a ds:Paper ;
         ds:hasTitle ?paperTitle ;
         ds:hasTopic ?localTopic .

  # 3. Topic connected to Wikidata
  ?localTopic a ds:Topic ;
              rdfs:label ?topicLabel ;
              owl:sameAs ?wikidataTopic .

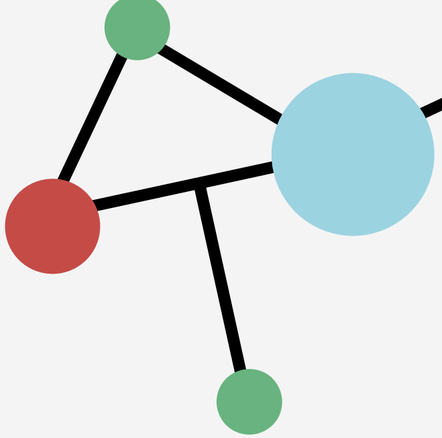
  # 4. Wikidata: topic is subclass of Mathematics
  SERVICE <https://query.wikidata.org/sparql> {
    ?wikidataTopic wdt:P279* wd:Q395 .
  }
}
```

Results

Author	Name	PaperTitle	TopicLabel
<http://dsgraph.org/ds/0000-0002-6331-7211>	"Fulvia Pennoni"	"A causal hidden Markov model for assessing effects of multiple direct mail campaigns"	"Statistics and Probability"
<http://dsgraph.org/ds/0000-0003-0606-5852>	"Paola Rebora"	"Sampling strategies to evaluate the prognostic value of a new biomarker on a time-to-event end-point"	"Statistics and Probability"
<http://dsgraph.org/ds/0000-0003-0606-5852>	"Paola Rebora"	"Correction: Sampling strategies to evaluate the prognostic value of a new biomarker on a time-to-event end-point"	"Statistics and Probability"
<http://dsgraph.org/ds/0000-0001-9655-7142>	"Davide Chicco"	"The coefficient of determination R-squared is more informative than SMAPE, MAE, MAPE, MSE and RMSE in regression analysis evaluation"	"Statistics and Probability"
<http://dsgraph.org/ds/0000-0001-8771-5428>	"Davide Paolo Bernasconi"	"Modeling the hazard of transition into the absorbing state in the illness-death model"	"Statistics and Probability"
<http://dsgraph.org/ds/0000-0003-0606-5852>	"Paola Rebora"	"Modeling the hazard of transition into the absorbing state in the illness-death model"	"Statistics and Probability"
<http://dsgraph.org/ds/0000-0002-6624-8637>	"Marco Fattore"	"A fuzzy posetic toolbox for multi-criteria evaluation on ordinal data systems"	"Computational Theory and Mathematics"
<http://dsgraph.org/ds/0000-0002-8987-100X>	"Elisabetta Fersini"	"Safe Optimal Control of Dynamic Systems: Learning from Experts and Safely Exploring New Policies"	"Computational Theory and Mathematics"
<http://dsgraph.org/ds/0000-0002-4062-0824>	"Enza Messina"	"Safe Optimal Control of Dynamic Systems: Learning from Experts and Safely Exploring New Policies"	"Computational Theory and Mathematics"

Data Querying

Using Reasoner to make Inference on Chain Rule



Topic Insights

Which authors from the Data Science Course at the University of Milano-Bicocca have an interest in the topic 'General Economics, Econometrics and Finance'?

```
PREFIX ds: <http://dsgraph.org/ds#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX xsd: <http://www.w3.org/2001/XMLSchema#>

SELECT DISTINCT ?author ?authorName
WHERE {
 ?author ds:hasInterest ?topic ;
 ds:hasDSDepartment ?dept .

 FILTER(?dept = 1)

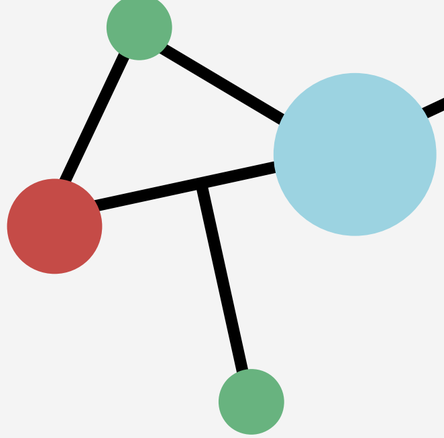
 ?topic rdfs:label ?topicLabel .
 FILTER(STR(?topicLabel) = "General Economics, Econometrics and
Finance")

 OPTIONAL { ?author rdfs:label ?authorName . }
}
```

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Question | **Query** | **Result**

Data Querying



Expand the Knowledge base by Federated query

```
PREFIX ds: <http://dsgraph.org/ds#>
PREFIX owl: <http://www.w3.org/2002/07/owl#>
PREFIX wdt: <http://www.wikidata.org/prop/direct/>
PREFIX p: <http://www.wikidata.org/prop/>
PREFIX ps: <http://www.wikidata.org/prop/statement/>
PREFIX pq: <http://www.wikidata.org/prop/qualifier/>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX wikibase: <http://wikiba.se/ontology#>
PREFIX bd: <http://www.bigdata.com/rdf#>

SELECT ?rank ?year ?rankingLabel WHERE {

  ?inst a ds:Institution ;
        ds:institutionName ?name ;
        owl:sameAs ?wikidata .
  FILTER(CONTAINS(LCASE(STR(?name)), "bicocca"))


  SERVICE <https://query.wikidata.org/sparql> {
    ?wikidata p:P1352 ?rankStatement .
    ?rankStatement ps:P1352 ?rank .
    ?rankStatement pq:P585 ?year .
    ?rankStatement pq:P459 ?rankingMethod .

    ?rankingMethod rdfs:label ?rankingLabel .
    FILTER(LANG(?rankingLabel) = "en")
    FILTER(CONTAINS(LCASE(STR(?rankingLabel)), "qs"))
  }
}
ORDER BY DESC(?year)
```

Institution Insights

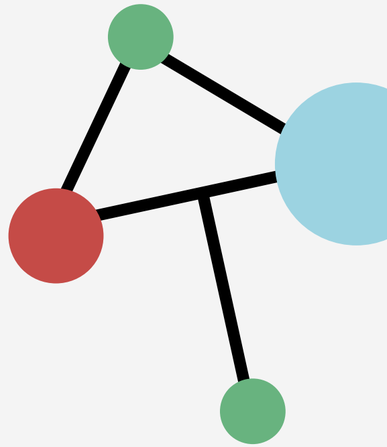
What QS rankings has the University of Bicocca received over the years?

Rank	Year	RankingLabel
513	2025-01-01T00:00:00Z	"QS World University Rankings"@en
481	2024-01-01T00:00:00Z	"QS World University Rankings"@en
450	2022-01-01T00:00:00Z	"QS World University Rankings"@en

Query | **Question** | **Result**

Data Querying

Using Reasoner to make Inference on Chain Rule



Query

```
PREFIX ds: <http://dsgraph.org/ds#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-
schema#>

SELECT DISTINCT ?institution ?label
WHERE {
 ?author ds:fullName "Matteo Palmonari" .
 ?author ds:hasContactWith ?institution .
 OPTIONAL { ?institution rdfs:label ?label . }
 FILTER(STRSTARTS(STR(?institution),
"https://openalex.org/"))
}
```

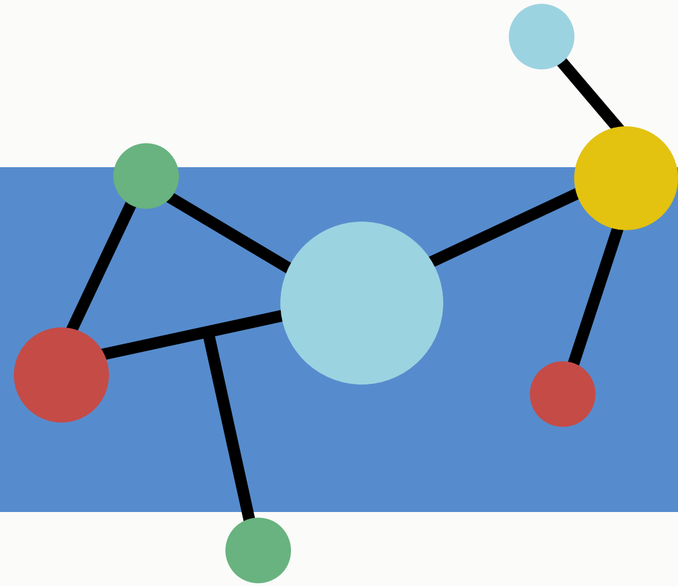
Which institutions is Prof. Palmonari in contact with?

Question

Results

institution	label
https://openalex.org/I15196421	University of Applied Sciences and Arts of Southern Switzerland
https://openalex.org/I3006985408	Jožef Stefan Institute
https://openalex.org/I46831270	Ontotext (Bulgaria)
https://openalex.org/I102248843	Vienna University of Economics and Business
https://openalex.org/I4210167034	Millennium Institute for Integrative Biology
https://openalex.org/I160993911	Queensland University of Technology
https://openalex.org/I165862685	St George's, University of London
https://openalex.org/I35440088	ETH Zurich
https://openalex.org/I4210117005	Laboratoire des Sciences du Numérique de Nantes
https://openalex.org/I4210107205	Russian State University for Physical Education Sport Youth and Tourism
https://openalex.org/I97018004	Stanford University
https://openalex.org/I66752286	University of Milano-Bicocca
https://openalex.org/I100066346	University of Stuttgart
https://openalex.org/I145847075	TU Wien
https://openalex.org/I62916508	Technical University of Munich
https://openalex.org/I173888879	SINTEF

Further developments



Expanding the Knowledge graph

Extend the ontology to include research assistants and postdocs from Bicocca’s computer science labs, allowing for a more complete and accurate representation of collaborations and citations in the department.

Improve word matching

Enhance topic matching by incorporating semantic similarity, enabling queries like software engineering to return conceptually related topics such as computer engineering, beyond exact keyword matches.

Enhancing User Experience with GraphRAG

Integrate a GraphRAG-based interface to allow users to ask natural language questions and receive context-aware answers grounded in the graph's inferred relations and semantic structure.